Skills Summary

Ratios and Equivalent Ratio Tables

Use this reference while completing your worksheet or when studying.

Writing a Ratio Three Ways

Rule: A ratio compares two quantities **in order**. The same ratio can be written three ways: with the word "to" (3 to 5), with a colon (3:5), or as a fraction $\left(\frac{3}{5}\right)$. The quantity named first always goes first.

Example: A team wins 4 games and loses 9. Write the ratio of wins to losses three ways.
Step 1. Wins are named first, so the win count goes first in every form.
Step 2. 4 wins to 9 losses: 4 to 9, 4:9, $\frac{4}{9}$ — three ways to write $\mathbf{\frac{4}{9}}$.

Try it: A team wins 5 games and loses 7. Write the ratio of wins to losses as a fraction.

check: $\frac{5}{7}$

(!) Watch out: Order matters! The ratio of dogs to cats is **not** the same as the ratio of cats to dogs — flipping the order names a different comparison.

Part-to-Part vs. Part-to-Whole

Rule: A **part-to-part** ratio compares one group to another group. A **part-to-whole** ratio compares one group to the total. To find the whole, **add the parts first**.

Example: A box holds 3 red pens and 5 black pens. Write the ratio of red pens to **all** pens.
Step 1. "All pens" means the whole, so add the parts first: $3 + 5 = 8$ pens.
Step 2. Red to all is part-to-whole: 3 out of 8, so the ratio is $\mathbf{\frac{3}{8}}$.

Try it: A jar holds 2 green buttons and 7 yellow buttons. Write the ratio of green buttons to **all** buttons.

check: $\frac{2}{9}$

Making and Checking Equivalent Ratios

Rule: Equivalent ratios name the same comparison. Multiply or divide **both** parts of a ratio by the same nonzero number to get an equivalent ratio. Two ratios are equivalent when they simplify to the same simplest form.

Example: Find the missing value: $\dfrac{5}{8} = \dfrac{?}{24}$.
Step 1. Find what the denominator was multiplied by: $8 \times 3 = 24$, so both parts must be multiplied by 3.
Step 2. Numerator: $5 \times 3 = 15$, so the missing value is **15**.

Try it: Find the missing value: $\dfrac{2}{5} = \dfrac{?}{30}$.

check: **12**

Key Vocabulary

Ratio: a comparison of two quantities, in order

Part-to-whole: a ratio comparing one group to the total

Equivalent ratios: ratios that name the same comparison

Simplest form: both parts share no common factor but 1